

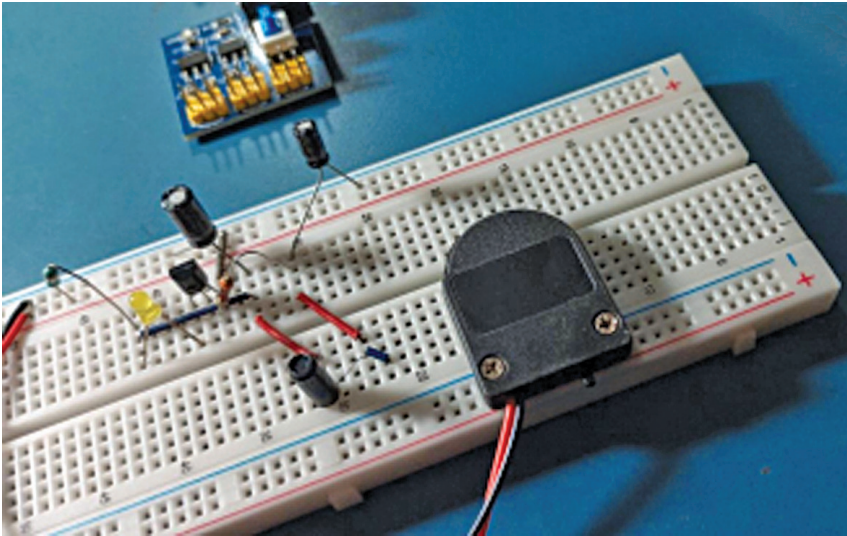


Fig. 6: Author's prototype on breadboard

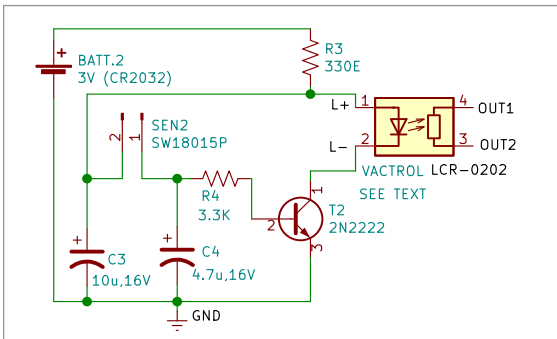


Fig. 7: Expansion circuit

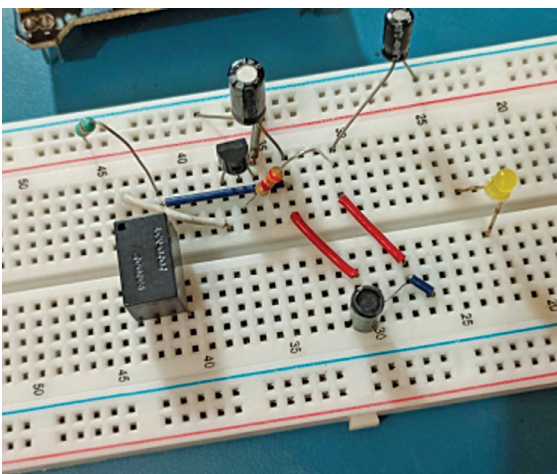


Fig. 8: Prototype of expansion circuit

custom-designed printed circuit board (PCB) is preferable, as it gives the device the charm of a

commercial module. The layout of the components is not very critical. Follow Fig. 5 for proper mounting of the vibration sensor switch.

After successful assembly, test the device by tapping it with a rubber mallet and ensure that the LED blinks briefly in response. The author's prototype on a breadboard is shown in Fig. 6. Now, the standalone shock and vibration sensor device can detect the magnitude of the shock and/or vibration and provide a visual response when its amplitude exceeds a certain threshold, defined by the sensor element itself.

Expansion

By using a few more components, this device can be turned into a fancy lamp that reacts to shock and vibration in its vicinity.

PARTS LIST (EXCLUDING EXPANSION CIRCUIT)

Semiconductors:

- T1 - 2N2222 NPN transistor
- LED1 - 5mm yellow LED

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 330-ohm
- R2 - 3.3-kilo-ohm

Capacitors:

- C1 - 10μF, 16V electrolytic
- C2 - 4.7μF, 16V electrolytic

Miscellaneous:

- Batt.1 - CR2032 3V coin cell
- SEN1 - SW18015P vibration sensor switch (or equivalent)

Ultimately, the device produces a unique negative-going pulse at its output.

Expansion tips. A minor drawback of this simple design is that the output pulse pattern cannot be evaluated for further processing. Fortunately, an easy solution is to replace the existing LED with a Vactrol (resistive opto-isolator), as shown in Fig. 7.

The LCR0202 Vactrol is used in the breadboard prototype shown in Fig. 8. However, any other commercial equivalent or a homemade Vactrol can also be used here.

For reference, the datasheet for the LCR-0202 can be found at <http://pliki.aksotronik.pl/lcr0202.pdf>.

A Vactrol consists of two components enclosed in one package: a light-emitting diode (LED) and a photoresistor. Applying a voltage to the LED causes current to pass through it and emit light, which strikes the photoresistor. If a voltage is applied to the photoresistor, it will generate a current proportional to the voltage applied to the LED. This makes the photoresistor a voltage-controlled resistor (VCR). The author's prototype of the expansion circuit is shown in Fig. 8. **EFY**



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